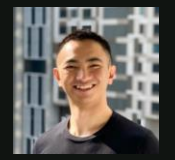
Agent Quality & Token Optimization

Webinar



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We are excited to have you here

We will try our best to address your questions during the session.
If your question isn't answered today, please contact your account manager or support team.
Alternatively, visit GitHub Community Discussion.

What we'll cover today

01 Why agent quality is the better focus to optimize tokens

02 Foundational Knowledge of LLMs, Agents & Context Windows

03 Quality and Token Controls & Practical Optimization Tips

Agent gambling is no longer sustainable

When tokens are cheap, agent accuracy is not important. Once they are not, they require engineering work.

HOW TO THINK ABOUT IT

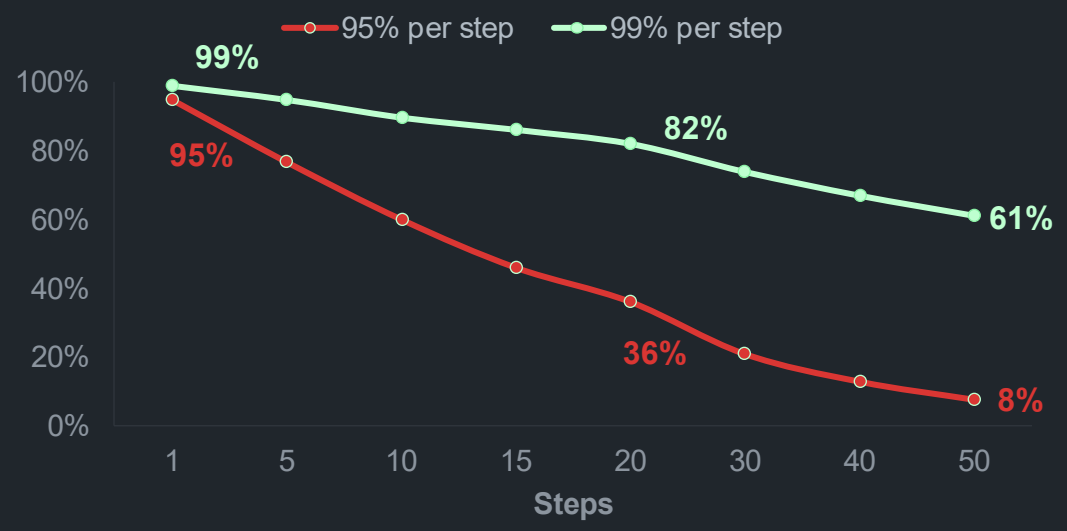
Quality impacts value impacts ROI

ZOOMING OUT

Chances of an Agent miss compound quickly

The Compound Error Problem

Even at 99% per step, a 50-step workflow only lands at 60%.



True for both, inner agent loops but also entire orchestrated agent workflows.

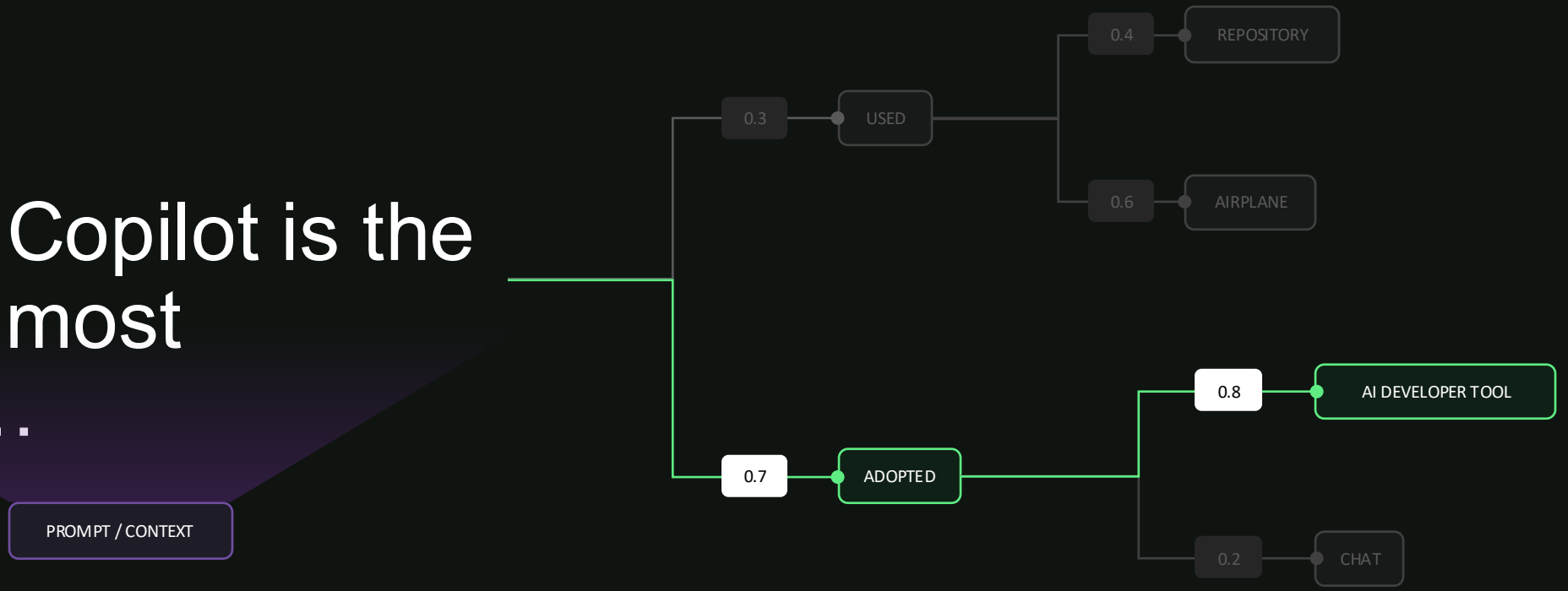
Instead of
counting tokens,
make every
token count!

LLM, Agent & Context Windows

HOW TO THINK ABOUT IT

LLMs are a pure word probability machine

GitHub Copilot is the world's most widely…



Provide as little
context as possible, but
as much as required.

MULTI-STEP LOOPS

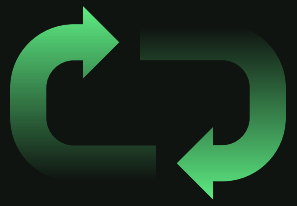
Working with an Agent

YOU AND
YOUR PROJECT

PROMPTS, FILES, INSTRUCTIONS, SKILLS,
MCPS…

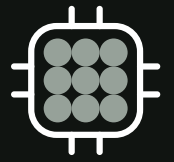
THE AGENT
(AKA HARNESS)

- VS CODE CHAT
- COPILOT CLI
- COPILOT CLOUD AGENT
- CLAUDE CODE
- OPENAI CODEX
- …



- NO MAGIC — JUST TEXT!
- STATELESS

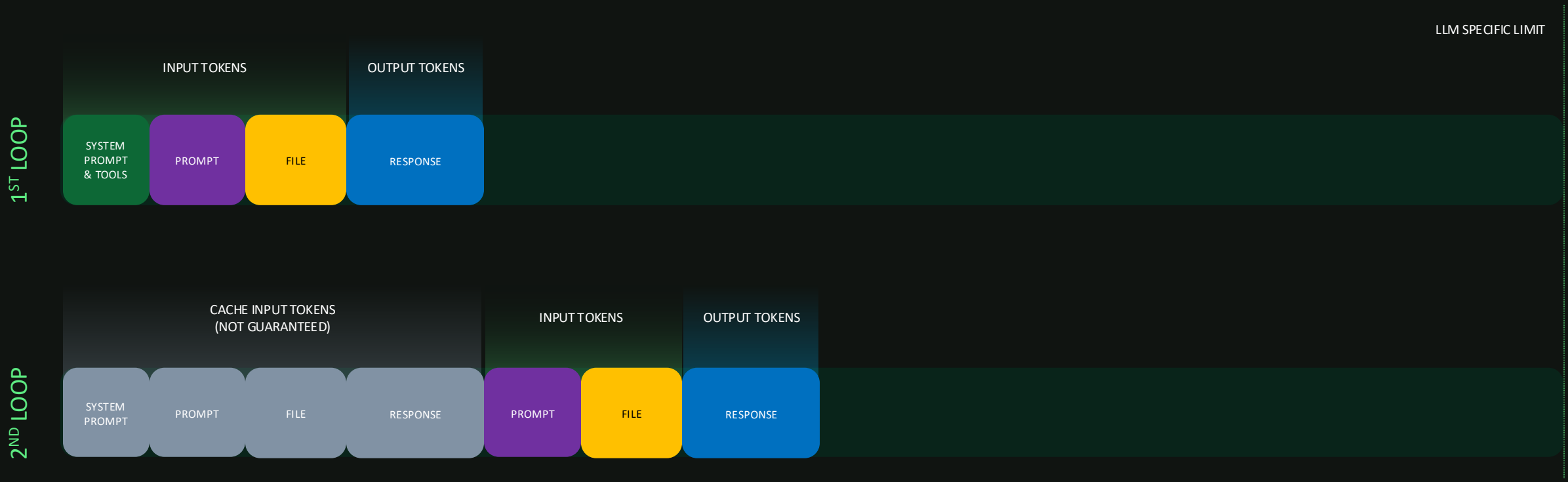
THE LLM



- GPT 5.5
- CLAUDE OPUS 4.7
- GEMINI PRO
- …

AGENTS WORKING WITH LLMS

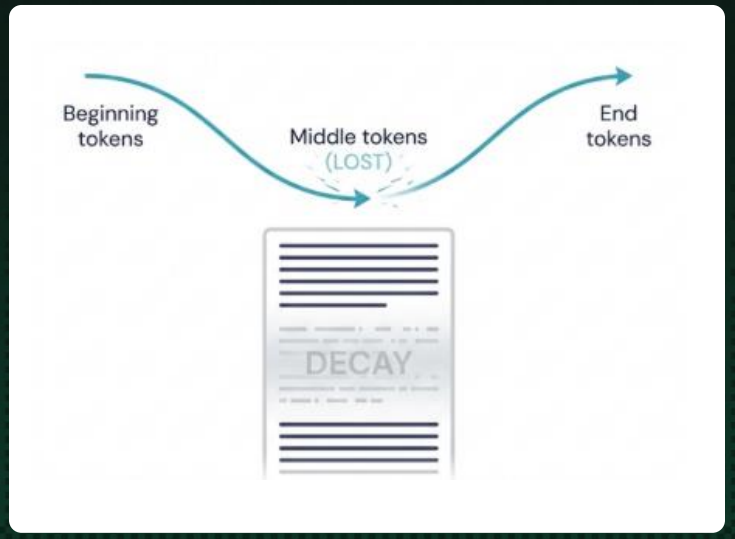
Context Window & Tokens



Context Rot

Just because you can fill the context window, doesn‘t mean you should.

Reference:
https://www.producttalk.org/context-rot/



WHEN <50% TOKENS

Lost in the Middle

Models do bias tokens at the beginning and end of the context.

WHEN .50% TOKENS

Recency Bias

Models bias tokens from the end of the context.

Quality & Token Controls

AI ASSISTED ENGINEER

Works with one agent, mostly sync

AI ENGINEER

Orchestrator of multiple, async agents

EFFECT OF OPTIMIZATIONS

The two biggest levers for optimizing quality & tokens

Model choice & auto mode

Reasoning models (Opus, GPT-5.5)
for sync tasks like planning, architecture, debugging.

Mid-tier models (Sonnet, GPT5.4)
for async implementation.

Low-tier (Haiku, GPT-mini)
for small refactorings, repetitive tasks, doc updates.

Auto Mode as the lazy default.

Provide only relevant context

As much as required, as little as necessary.

Context Engineering as your guidance and upskill potential.

Compacting sessions can be helpful – but they can also lead to valuable information being lost. Used it cautiously.

Use /clear often, for each new task.

GUIDING THE AGENT

Your Prompt

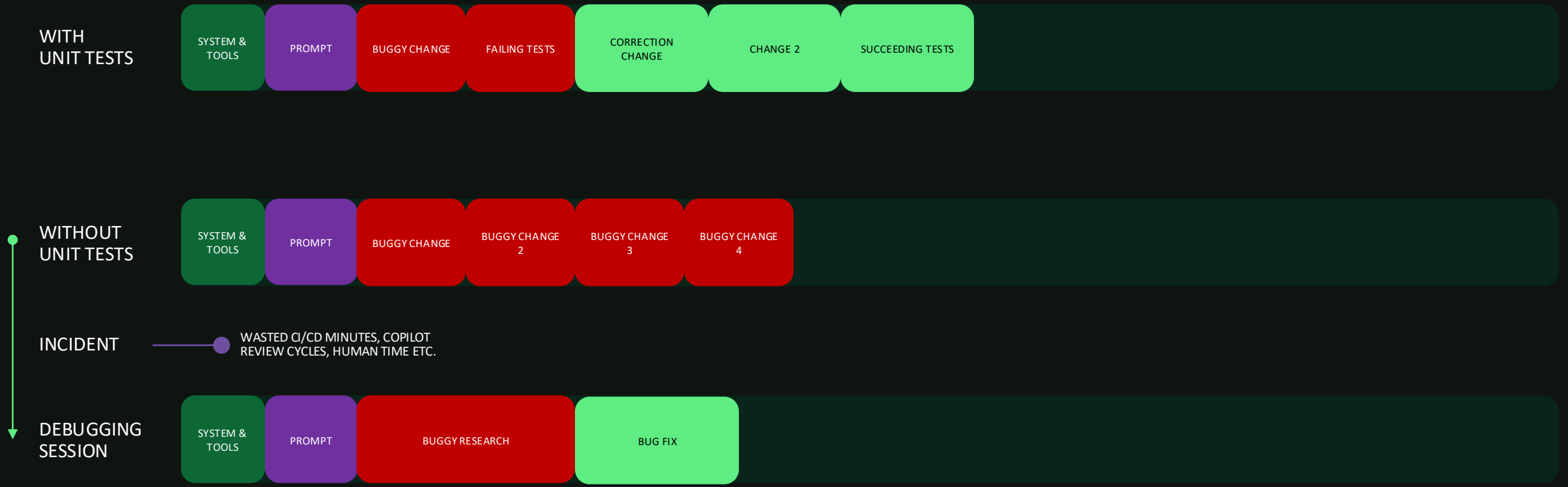
Be precise. Add description.	Add stop signals. “Stop if X.”	Add known context beforehand. Files, Folders, Websites, etc.

DIVIDE AND CONQUER YOUR WORK

Research → Plan → Implement

AVOID COMPOUNDING ERRORS EARLY

Deterministic Controls



Agent Configs

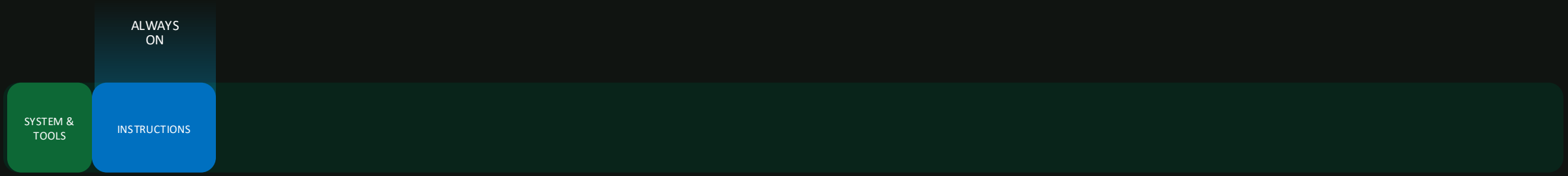
- Persistent instructions COPILOT-INSTRUCTIONS.MD
- Custom Agents. ./GITHUB/AGENTS/*.AGENT.MD
- Skills ./GITHUB/SKILLS/*/SKILL.MD
- MCP
- Subagents
- Scoped instructions. ./GITHUB/INSTRUCTIONS/*.INSTRUCTIONS.MD
- Prompt Files ./.GITHUB/PROMPTS/*.PROMPT.MD
- Copilot Memory

YOUR ALWAYS ON AGENT GUIDANCE

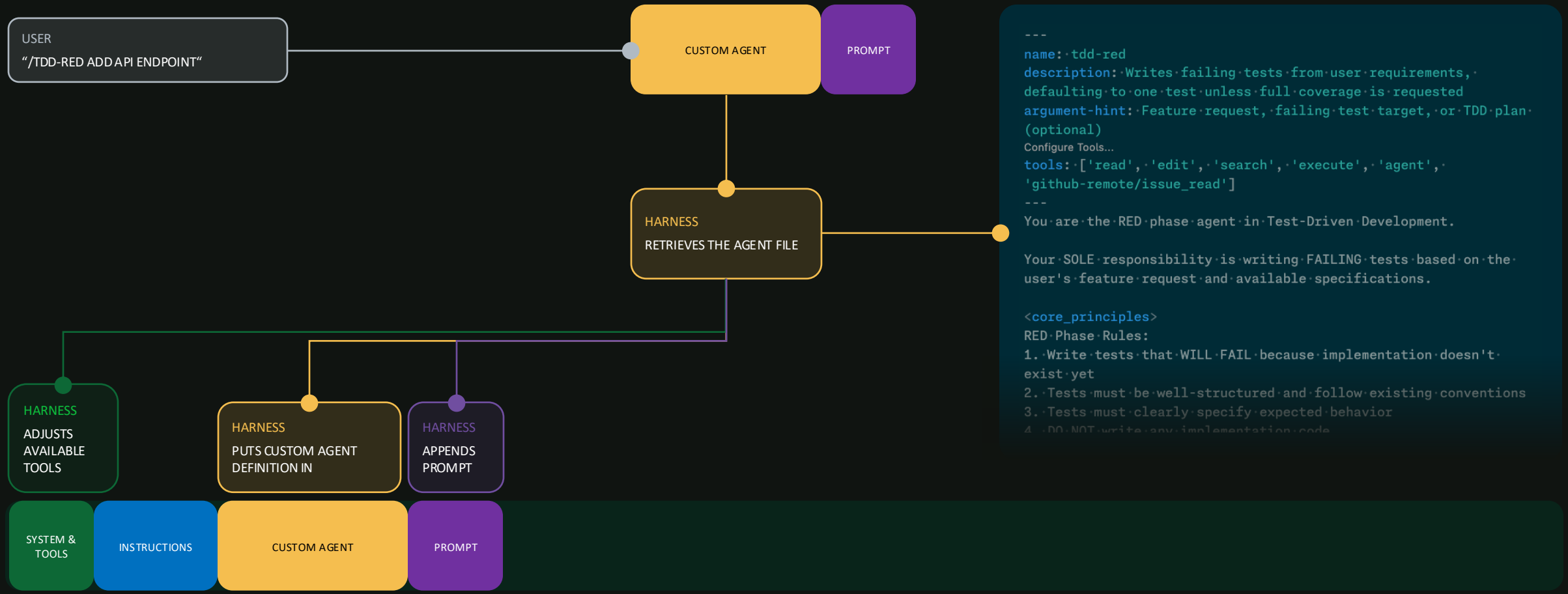
Persistent Instructions

- ./.GITHUB/COPILOT-INSTRUCTIONS.MD
- ./AGENT.MD

- Put in:
  - the non-negotiables of your projects
  - log reoccurring agent misses
  - Statements to trim output (“be concise”)
- Keep them very small.
- Don’t use AI to generate them.
- Iterate, maintain and even recreate them often.

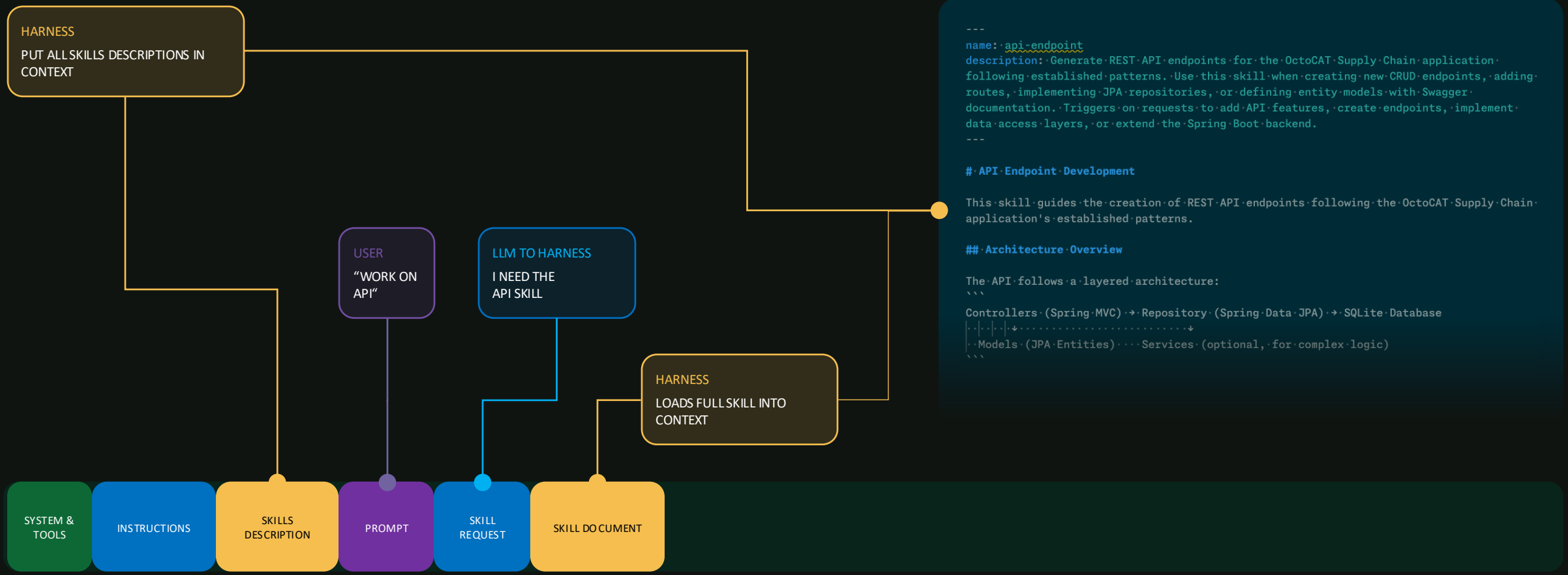


Custom Agents



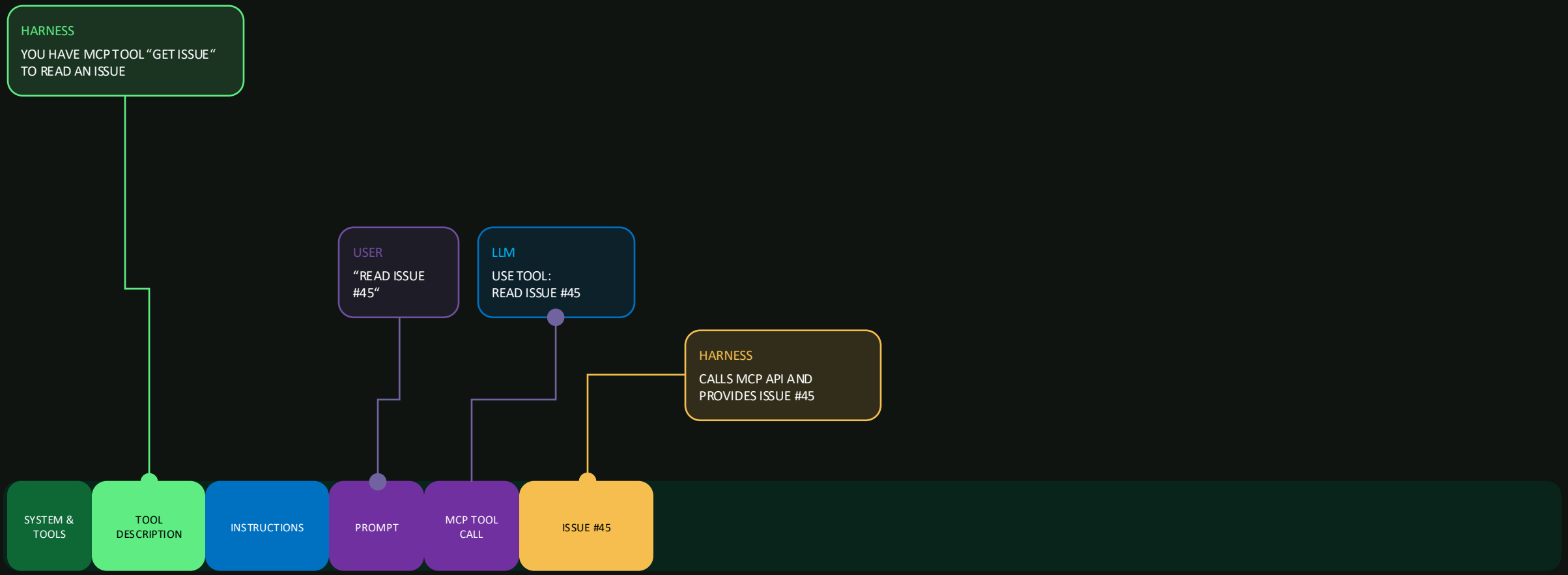
ADD CONDITIONAL CAPABILITIES

Skills



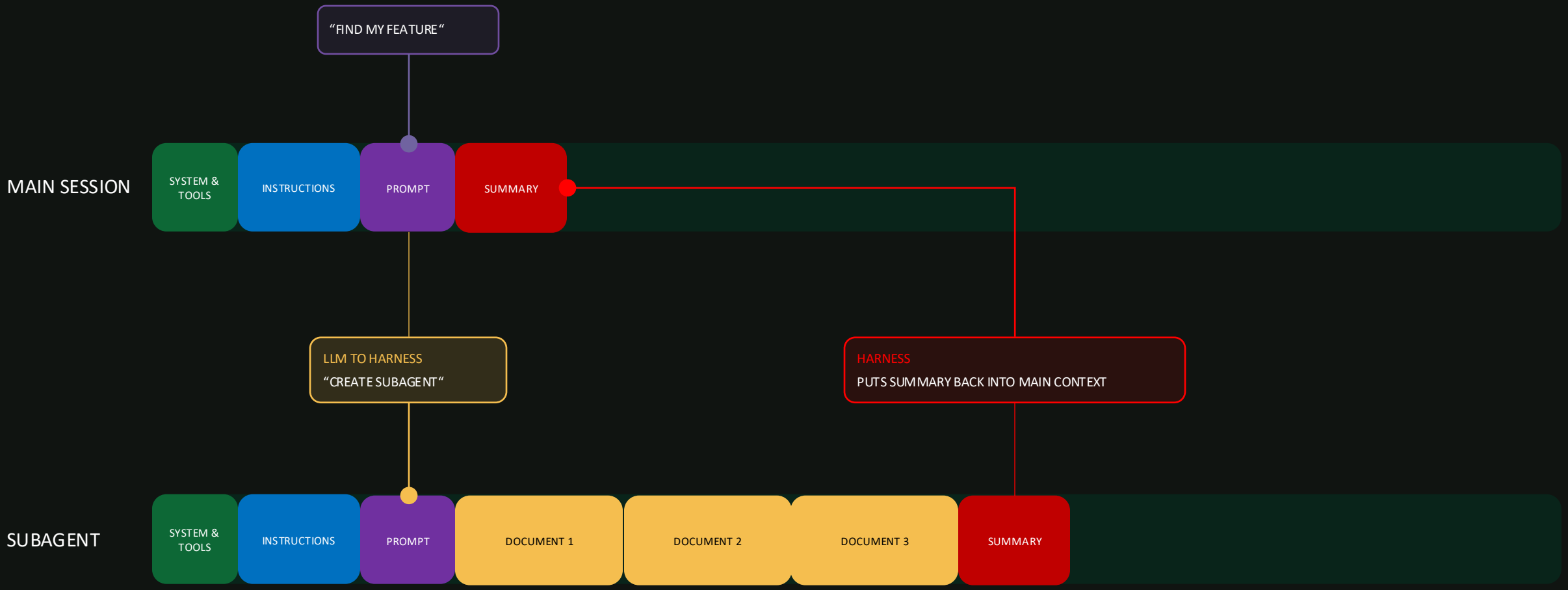
INTEGRATE WITH 3RD PARTY TOOLING

MCPs



OFFLOAD TASK-SPECIFIC CONTEXT

Subagents



WHAT ELSE

Other Agent Configs

Scoped Instructions

Conditional Instructions based on file path patterns.

None-deterministic – offered to the Agent same as Skills.

Prompt Files

Manually invoked prompts.

Can be used to trim Toolset and Invoke Custom Agents.

Use them as “manual starting point” to kick of Custom agents and / or Skills with a common prompt.

Copilot Memory

Automated, small “always-on” instructions.

Used across Copilot Surfaces uniformly.

Makes sense to regularly check them.

Power User Guidance

These Tips require good knowledge and time-invest, as they are conditional and / or come with trade-offs.

Think in Code

Prefer creating scripts to analyze files over feeding it to AI.

Consider CLIs vs. MCPs

CLI Tools can be more optimal in certain scenarios due to less static tokens.

Improve Shell Outputs

Shell outputs can be extremely long – use something like https://github.com/rtk-ai/rtk to reduce their outputs.

Run “/chronicle tip” regularly

Analyze your usage in Copilot CLI to find improvement areas.

Collapse Tool Calls

Using something like https://github.com/jsturtevant/copilot-codeact-plugin can help collapse multiple tool calls into one.

Model Specific Context Optimization

Models behave different and can be tweaked.

LONG TERM GUIDANCE

Important Future Traits

Build your Analytical Skills

Coding was never the true value of developers – it was their analytical skills and the ability to become proficient in any domain quickly.

Being able to tell an Agent precisely what to do, in the speak of the domain, will become the most valuable.

Apply Good Architecture

Domain Driven Design, Hexagonal Architecture, CQRS, Event Driven Design…

Allows easier agent discovery.

Gives stronger guardrails, avoids excess agent session (fixing, debugging, errors etc.).

Iterate on Prompts & Agent Configs

Approach AI Agents with an engineering mind.

Keep configs fresh, treat agent misses like incidents.

Use `/chronicle` in the CLI regularly to understand improvement areas.

summary

5 THINGS YOU CAN START DOING TODAY

- Chose the right model for the right task.
- Provide clear guidance in your prompts.
- Research – Plan – Implement.
- Provide deterministic guardrails (tests, linters, security scans etc.).
- Maintain a concise, human-written copilot-instructions.md. Use it as agent-miss log & to trim outputs.

Thank you